

Steering Amine-CO₂ Chemistry: A Molecular Insight into the Amino Site Relationship of Carbamate and Protonated Amine

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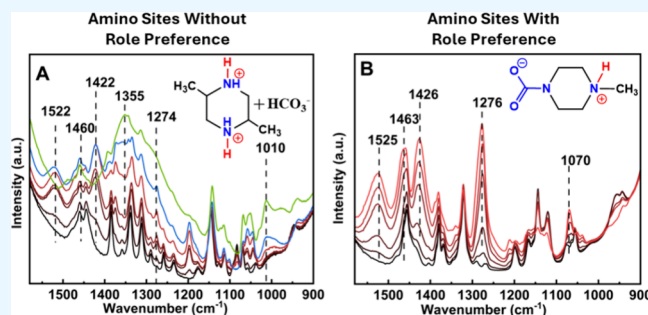


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ABSTRACT: The chemical absorption of carbon dioxide (CO₂) using amine solvents is set to be a prominent technology for large-scale point source carbon capture, yet its development encounters significant challenges, such as the inherent instability of carbamate intermediates that are crucial to the CO₂ capture process. Traditionally, amine solvent development has been focusing on stringent process controls and optimizing the structure of amino sites that form carbamates. Our study introduces an innovative concept of intersite stability, which enhances the relative structural arrangement and positioning between amino sites, specifically between carbamate and its counter-protonated amine. Our approach not only stabilizes the carbamate itself but also leverages synergistic effects to improve overall system performance under higher CO₂ loadings. We demonstrate that strategic increases in structural differences between amino sites can significantly augment stability, effectively mitigating the traditional pathways of carbamate decomposition to bicarbonate, especially under high-temperature CO₂ loading conditions. We optimize the amine system's resilience against decomposition by designating specific roles for amino sites, increasing protonation and carbamate stabilization, and strategically modifying their structural relationships, both within and between molecules. Our detailed analysis and validation of this concept includes the study of structural positioning in blends of cyclic amines, multialkylamines, and alkanolamines, which are candidates for commercialization. These modifications made to traditional amine systems, which usually form less stable carbamates, follow the logic of our intersite stability framework, showcasing enhanced stability even in aqueous environments. This study paves the way for more reliable and efficient CO₂ capture technologies by fundamentally rethinking the dynamics within the amino sites of these amine-based solvents.



INTRODUCTION

The quest for effective solvents for carbon capture is increasingly essential in the fight against climate change. As the search for these solvents evolves, a primary focus has been on reducing regeneration energy.^{1–11} However, the energy efficiency should not be the sole criterion for solvent selection. Equally critical is the propensity of carbamate products to decompose into bicarbonate under specific conditions—a process that directly impacts the overall efficacy of the CO₂ capture process. The simultaneous presence of carbamate and bicarbonate in CO₂ capture systems introduces significant operational challenges due to their differing reaction kinetics and environmental requirements. Carbamates form rapidly in environments with a low water content, facilitating efficient CO₂ absorption. In contrast, the formation of bicarbonates depends on the presence of water and proceeds through slower, multistep reactions.^{12–17} This divergence in the process conditions complicates the simultaneous optimization of the system for both reaction pathways. Moreover, the presence of species with varying thermal stabilities escalates the energy demands for solvent regeneration, further complicating efforts to enhance energy efficiency. Preventing the decomposition of

carbamate to bicarbonate is thus crucial for improving CO₂ capture efficiency, maintaining system performance, and reducing operational costs. To manage these challenges, adjustments in operating conditions such as gas flow rates and temperatures have been implemented.^{1,18,19} However, these process-based mitigation strategies encounter significant limitations when scaling up to capture millions of tons of CO₂.^{1,3,9,10} Consequently, a deep understanding and mitigation of the reaction pathways for the amine decomposition are imperative for enhancing the efficiency of CO₂ capture technologies with a concentrated focus on developing stable absorption product chemistries.

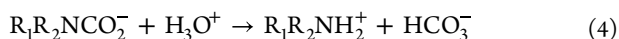
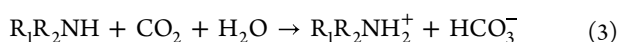
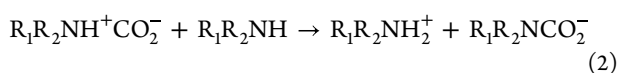
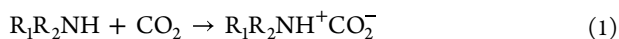
Various strategies for solvent discovery have focused on novel formulations of amine-based blends, including single and

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multicomponent alkanolamines, cyclic amines, alkyl carbonate amines, and superbase amines, all aimed at enhancing CO₂ capacity, reaction kinetics, and other physicochemical properties.^{20,21} Besides, some other work dedicated effort to harvest the cooperative effects in the form of carbamic acids on either solid material^{22–24} or water lean amine systems^{25–28} to accommodate a high ratio of CO₂ capture. While the intra- and intermolecular interaction of carbamate has been plentifully discussed in the open literature, these discussions rarely focus on the molecular structures controlling the relationship between the reaction mechanism and stability of carbamates that are formed during CO₂ absorption, especially in aqueous systems. It has been widely accepted that the two primary pathways for the reaction between amine solvent and carbon dioxide involve the formation of carbamate and bicarbonate.^{12,14,29,30}



in which R_1R_2NH : an amine molecule; $R_1R_2NH^+CO_2^-$: a zwitterion formed from the reaction of amine and CO₂; $R_1R_2NH_2^+$: a protonated amine formed from the binding of a proton and an amino site; and $R_1R_2NCO_2^-$: a carbamate moiety formed from the anchoring of CO₂ and an amino site. Importantly, these are competing pathways. On one side, carbamate formation begins when the amino site of an amine molecule attacks the electrophilic carbon of CO₂, leading to the formation of a zwitterion intermediate (eq 1). This zwitterion then transfers a proton to the amino site of another unreacted amine, leading to the carbamate formation (eq 2). The subsequent hydrolysis or decomposition of carbamate in the presence of water yields bicarbonate and free amine (eq 4). On the other hand, CO₂ can be directly hydrolyzed to bicarbonate in the presence of an amine acting as a base catalyst (eq 3). While the formation of bicarbonates, through either mentioned pathway, is slower compared to the carbamate formation step,^{16,17,31} bicarbonates allow for higher CO₂ loading than carbamates.

To develop high-performance absorbents, a comprehensive mechanistic understanding of the specific reaction pathways of carbamate hydrolysis (eq 4) and the direct hydrolysis of CO₂ to bicarbonate (eq 3) is essential. It has been observed that amines with different structures exhibit varying degrees of carbamate hydrolysis.^{32,33} Certain amines predominantly form carbamates at low CO₂ loadings but undergo hydrolysis to produce bicarbonates when their carbamates become voluminous under high CO₂ loadings. Insight into the impact of diverse structural variations on the stability of carbamate across different amine solvents, along with a detailed understanding of the elementary reactions in carbamate hydrolysis, is gaining increasing attention.^{34–38} Specifically, experimental studies have highlighted that sterically hindered alkanolamines, derived by varying methylation, chain length, and positioning of substituents near the amino site or nitrogen atom, influence the propensity of carbamate or bicarbonate formation.^{39–42} Additionally, computational studies on some simple alkanolamines, such as MEA and AMP, use reaction pathway

modeling and investigate the intermediate state of the water-carbamic acid complex and proton transfer from zwitterion to a nearby neutral amine.^{12,14,30,43–46} Despite these advances, the underlying mechanisms that dictate the preferential formation of carbamate versus bicarbonate remain fragmented and incomplete, representing a critical area for further research.

In our research, we explored how site preference for carbamates and protonated amines—usually referred to as conjugate acids of the amines—influences their stability. We introduce the concept of “intersite stability” to elucidate the stability mechanisms of carbamate. This concept offers a coherent interpretation based on the chemical structure of the molecule, focusing on the relationship between carbamate decomposition and the arrangement of amino sites and their preferences to form carbamate versus protonated amine. This fundamental mechanistic understanding of intersite stability has been validated through both experiments and computational modeling. Moreover, we demonstrated practical applications of this concept by illustrating how amine systems with previously unstable carbamates can attain enhanced product stability through modifications based on the principles of intersite stability.

METHODOLOGY

Experimental Method. Amines, including 2,5-dimethylpiperazine, 2,6-dimethylpiperazine, 1-methylpiperazine, piperidine, 2-methylpiperidine, morpholine, monoethanol amine (MEA), and aminomethyl propanol, are purchased from Sigma Aldrich as research-grade chemicals. Aqueous amine solutions are prepared by diluting neat amine in deionized water at a concentration of 10–20 mol % in 20 mL glass vials. The CO₂ capture reaction is conducted by bubbling a pure stream of CO₂ through vials filled with amine solution. The solution, once reacted, is sampled using a micropipette and analyzed using a Bruker Vertex 70 Fourier Transform Infrared (FTIR) spectrometer in an attenuated total reflection (ATR) mode. The FTIR spectra are collected with 1 cm^{−1} resolution and 40 accumulations ranging from 4000 to 850 cm^{−1}.

Computational Method. We used ORCA to conduct all the quantum chemistry calculations.⁴⁷ The crucial task to reveal the reaction mechanism involved in the CO₂ capture was to determine the minimum energy pathway (MEP) between the reactant and product states. The MEP is defined as the lowest energy path from two thermodynamically stable minima (i.e., reactant and product) with a transition state (TS) at the peak of its potential energy surface.^{48,49} The nudged elastic band (NEB) is a popular method to determine the MEP and TS, which can then calculate the reaction energy and activation energy.^{50–52} NEB first generates interpolated atomic positions between the reactant and product of a reaction. Each frame is connected by an imaginary elastic band that alters the atomic positions with spring forces perpendicular to the tangent between each frame. After optimization, the atomic positions along the reaction coordinate are optimized to the MEP, and the first-order saddle point is defined as the TS. In this work, we used ORCA to implement climbing image nudged elastic band (CI-NEB) optimization to improve the TS convergence.⁵³ In each simulation, one water molecule was present as an explicit solvent, and a conductor-like polarizable continuum model (CPCM) was implemented as an implicit aqueous solvation model^{54,55} to balance between accurate polar interactions and computational cost. In our workflow, the reactant and product were optimized with the r2SCAN-3c

functional as a diffusion and counterpoise-corrected triple- ζ Gaussian basis set.^{56–58} The NEB was conducted with the same method as the optimization, and the final energy was calculated at a more accurate level at wB97X-V/def2-TZVP with van der Waals correction.^{59–61} The reaction energy is defined as the energy difference between the product and reactant; the activation energy or energy barrier is defined as the energy difference between the TS and reactant.

RESULTS AND DISCUSSION

Intersites Stability of Protonated and Carbamate Sites. Focusing on the first reaction pathway, which involves three consecutive reactions 1, 2, and 4, where carbamate forms and subsequently decomposes, extensive studies suggest that the initial step of carbamate hydrolysis is the recombination of a proton from a protonated amine with the carbamate site to form carbamic acid, facilitated by the presence of water.^{14,29,30,62,63} This step occurs in the presence of water and is independent of whether the protonated site and carbamate site originate from the same or different molecules. From this perspective, the critical factor in carbamate decomposition is the ability to transfer protons and recombine the carbamate site with the protonated site. This implies that the stability of the carbamate product depends not solely on the absolute stability of the carbamate ion itself but rather on the overall collective stability of the carbamate site in conjunction with the adjacent protonated amine. Building on this rationale, we introduce the concept of "intersite stability" of carbamate and protonated sites as a key determinant of carbamate product stability. In a formal way, intersite stability in the context of carbon capture is the collaborative effect of two amino sites having distinct preferences in forming a protonated site versus a carbamate site, leading to the overall stability of the carbamate product. According to this concept, the stability of carbamate is intrinsically linked to the stability of the protonated amino site. The essence of the intersite stability, as depicted in Figure 1, is that if protonated sites are relatively stable in the presence of carbamate sites and vice versa, the final product of absorption will remain as a stable carbamate even in the presence of water or other forms of hydroxyl ions. Conversely, if either site type is not relatively stable in the presence of the other, the carbamate product will likely decompose into carbonate/bicarbonate. Practically, the intersite stability can be achieved by designing the amino sites so that specific molecules or sites are designated for fixed roles—one prone to protonation and the other prone to carbamate ion formation. This innovative concept of the intersite stability presents a significant opportunity to transform amine systems that typically form unstable carbamates into systems that exclusively produce stable carbamates. Through structural modulation, geometrical and chemical properties of protonated and carbamate sites, such as basicity, steric hindrance, and nucleophilicity, can be simultaneously tuned to collaboratively stabilize the carbamate products. It is worth noting that while most amine systems follow the zwitterion mechanism, a few, particularly smaller molecules like MEA, also participate in a termolecular reaction to form carbonate/bicarbonate.^{12,14,64} However, these amine systems are generally more conventional and have been thoroughly studied computationally and experimentally; thus, they are not the focus of this article.^{12,14,16,43,44}

This study investigates two primary scenarios demonstrating how intersite stability between carbamate and protonated sites

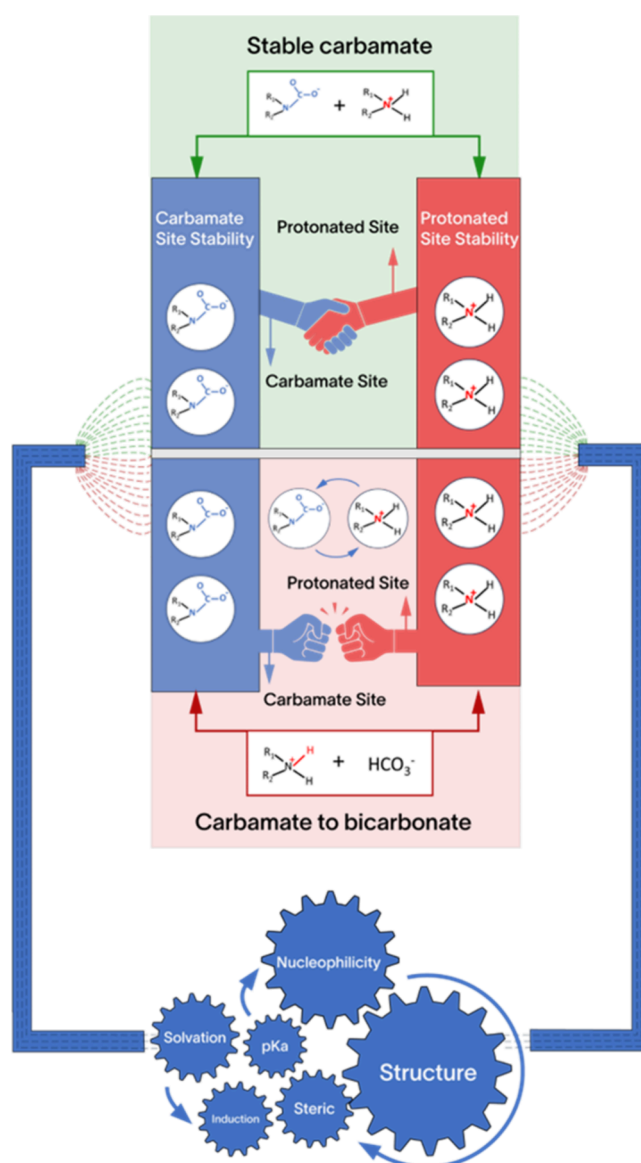


Figure 1. Schematic representation of the intersite stability concept in the perspective of carbamate and bicarbonate formation reaction pathways of CO₂ capture in amine-based solution. The top green box represents conditions where structural differences support amino sites with specific preferences for forming carbamates versus protonated amines—the core of the intersite stability concept. In this scenario, amino sites are optimally differentiated, enhancing the stability of carbamate and preventing its decomposition into bicarbonate. The bottom red box illustrates scenarios with minimal or no structural differences between amino sites. The lack of distinct site preferences in these cases can decrease the carbamate stability, thereby increasing the likelihood of its decomposition to bicarbonate. The gears symbolize the simultaneous collaboration among several drivers, including basicity, steric hindrance, and nucleophilicity, to achieve the structural differences necessary for establishing intersite stability.

can enhance the chemistry of amine solvents, particularly those that initially produce carbonates or bicarbonates at equilibrium conversion. In the first scenario, the induction of a stable carbamate product occurs through the intramolecular structure modulation, while the second scenario focuses on stability induced by intermolecular structure modulation.

Inducing Intersite Stability by Intramolecular Site Modulation. In the first scenario, we analyze an amine

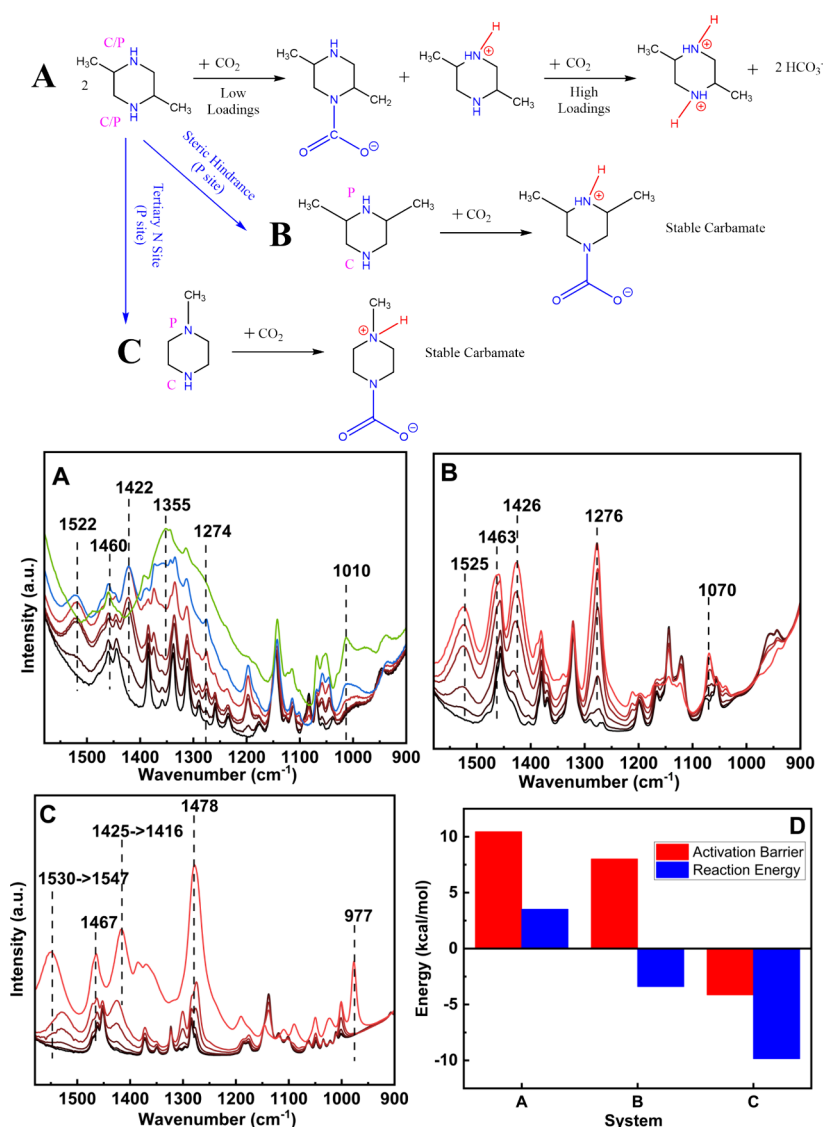


Figure 2. Intramolecular inter-site stability impact on the carbamate formation from piperazine derivatives. This figure illustrates the application of intersite stability principles in intramolecular systems, focusing on the formation and stability of carbamate in reactions involving different methyl-substituted piperazines. (A, B, and C) display the FTIR spectra for reactions involving 20 mol % of 2,5-dimethyl-piperazine, 2,6-dimethyl-piperazine, and 1-methyl-piperazine, respectively, tracking the progress from 0% conversion to equilibrium conversion. A highlight scenario with no structural differences between amino sites led to decreased carbamate stability and increased likelihood of decomposition to bicarbonate under high CO_2 loading. B and C demonstrate strategic increases in structural differences; B employs steric hindrance from a $-\text{CH}_3$ group at the α position to designate a preferred protonated site, while C utilizes a tertiary amino site to enhance protonation preference. These modifications significantly improve carbamate stability, preventing its decomposition into bicarbonate and illustrating the critical role of site differentiation in enhancing the efficacy of the CO_2 capture processes. (D) Activation energy and reaction energy of carbamate formation with methyl-substituted piperazine with carbamate binding to $\text{CH}_2\text{NHCHCH}_3$ of 2,5-dimethyl-piperazine (A), CH_2NHCH_2 of 2,6-dimethyl-piperazine (B), and CH_2NHCH_2 of 1-methyl-piperazine (C), respectively.

solution containing 2,5-dimethyl-piperazine dissolved in water as a diluent. At low amine conversions, the dominant product is carbamate. The FTIR spectra of this amine solution at low CO_2 loading exhibit key evolving bands around 1520, 1422, 1460, and 1274 cm^{-1} . These bands correspond to the vibrational modes of asymmetric and symmetric C=O stretching, N–H bending of NH_2^+ , and N–C stretching of N-COO^- , respectively. All these bands can be assigned to the formation of carbamate by 2,5-dimethyl-piperazine, as shown in Figure 2A. As the amine solution absorbs more CO_2 at higher amine conversion, the formed carbamate decomposes, evidenced by the disappearance of the C=O asymmetric band at 1520 cm^{-1} of carbamate. Concurrently, as the CO_2 loading

approaches equilibrium, the FTIR fingerprints of bicarbonate, including a broad band of HCO_3^- at 1355 cm^{-1} , and C–O stretching of HCO_3^- at 1010 cm^{-1} , become more enhanced. This behavior has been reported as the decomposition of carbamate to the bicarbonate product.^{32,33,65–67} In the view of the intersite stability concept presented in the previous section, the protonated site and carbamate site of 2,5-dimethyl-piperazine are structurally equivalent, in the form of $\text{CH}_2\text{NH}_2^+(\text{CHCH}_3)\text{CH}_2$ and $\text{CH}_2\text{NCOO}^-(\text{CHCH}_3)\text{CH}_2$, respectively. Both sites experience a steric effect from a methyl group at the α position. The structural equivalency of protonated and carbamate sites makes the carbamate product

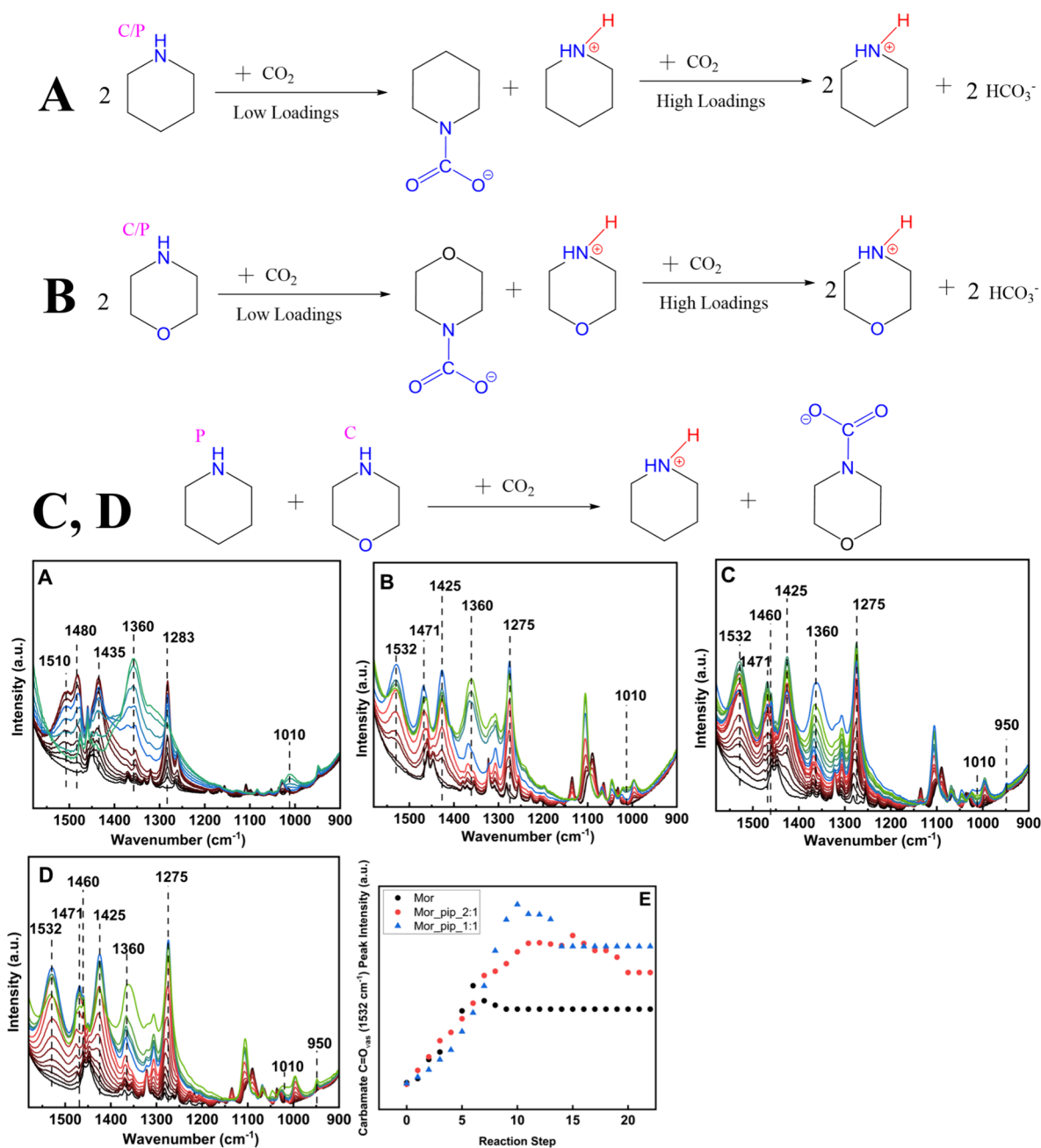


Figure 3. Application of intersite stability in intermolecular systems. (A–D) FTIR spectra of the CO₂ capture reaction with aqueous 20 mol % piperidine (A), aqueous 15 mol % morpholine (B), an aqueous mixture of 15 mol % morpholine and piperidine with the ratio of morpholine to piperidine to be 2:1 (C), and an aqueous mixture of 15 mol % morpholine and piperidine with the ratio of morpholine to piperidine to be 1:1 (D) from 0% conversion to equilibrium conversion. (E) Intensity of carbamate C=O_{vas} (1532 cm⁻¹) peak extracted from B, C, and D.

less stable. To prevent the decomposition of carbamate, the intersite stability of both sites must be improved.

There are two methods to modulate the structure intramolecularly in this case. The first method is to impose a greater steric hindrance on the protonated site while simultaneously decreasing the steric hindrance at the carbamate site. The modulation can simply be achieved by adding a methyl group to the α position for the protonated site and removing the methyl group at the α position for the carbamate site. This modification results in a blend of CH₃CHNHCHCH₃ and CH₂NHCH₂ sites in 2,6-dimethyl-piperazine solution. Such modification significantly enhances the preference of the CH₃CHNHCHCH₃ site to be protonated over the

CH₂NHCH₂ site, which is now preferable to become the carbamate site. The carbamate site preference can be predicted by DFT calculations. As shown in Figure 2D, the carbamate product from 2,6-dimethyl-piperazine, with proton binding to CH₃CHNHCHCH₃ and CO₂ binding to CH₂NHCH₂, has lower activation energy and negative reaction energy than the carbamate product from 2,5-dimethyl-piperazine. Thus, it is more spontaneous for CO₂ to readily bind to the CH₃CHNHCHCH₃ site, forming a zwitterion, which then transfers a proton to CH₂NHCH₂. To examine the impact of this modification on carbamate hydrolysis, we monitor the CO₂ absorbed by 2,6-dimethyl-piperazine solution using FTIR, and the spectra are shown in Figure 2B. The experimental

results show that at all CO₂ loadings, only carbamate is formed, indicated by the absence of two main bicarbonate FTIR features at approximately 1355 and 1010 cm⁻¹. This demonstrates that the stability of the carbamate product is enhanced, even in the presence of water, by simple methyl group modulation. It is significant to note that even at maximal amine conversion, 2,6-dimethyl-piperazine does not exhibit any signs of dicarbamate, whose N-COO⁻ band would appear around 1285 cm⁻¹; namely, one of the amino sites, the CH₃CHNHCHCH₃ site, is restricted from CO₂ binding. This observation further confirms that the specialization of amino sites by structural modulation is behind the stability of the carbamate product.

The second route to enhance carbamate stability through structural modification, starting with a solution of 2,5-dimethyl-piperazine, involves converting one amino site into a tertiary amine site, which will serve as the protonated site. Simultaneously, relocating a methyl group from the α position to the 1-position reduces steric hindrance at the secondary amine site, which will serve as the carbamate site. A study on CO₂ absorption by 1-methyl-piperazine is conducted as a showcase, the results of which are illustrated in Figure 2C. The FTIR spectra, recorded during incremental CO₂ absorption, exclusively display carbamate features, including bands at 1547, 1416, 1467, and 1278 cm⁻¹. These bands correspond to the vibrational modes of asymmetric and symmetric C=O stretching, N–H bending of NH₂⁺, and N–C stretching of N-COO⁻, respectively. This strategic structural modification also remarkably enhances the stability of the carbamate of the capture solution. In this application, for example, when the conversion is at a higher extent, the stretching band of C=O blue-shifts from 1530 to 1547 cm⁻¹ while a very strong band appears at 977 cm⁻¹, which is attributed to the C–N–H twisting mode. Although the roles of amino sites of 1-methyl-piperazine can be inferred from its chemical structure, as previously mentioned, theoretical analysis is necessary to confirm that each type of site fulfills its designated role. Theoretical considerations, as presented in Figure 2D, show that carbamate formation with CO₂ binding to CH₂NHCH₂ and proton binding to CH₃CHNHCHCH₃ of 2,5-dimethyl-piperazine or CH₃N(CH₂)₂ of 1-methyl-piperazine has significantly more negative reaction energy than carbamate formation from 2,5-dimethyl-piperazine solution. As a result, at equilibrium conversion, carbamate products become more preferentially formed, with the protonated moiety modified to be more sterically hindered. This confirmation supports the intersite concept, which emphasizes role-specific designation to enhance carbamate stability with intramolecular modulation in the capture solution.

Inducing Intersite Stability by Intermolecular Site Modulation. The formation of zwitterion and its subsequent proton transfer process happen spontaneously within collections of amino sites in the solution, regardless of the molecular origin of these sites. This means that the amino sites might come from the same molecule, in diamines or triamines, or from different molecules. This commonality in carbamate and bicarbonate formation pathways for both aqueous monoamines and diamines has been documented previously.^{36,68} Herein, we propose that the intersite stability can also be induced through intermolecular site modulation, essentially through physical mixing of sites. If two amine molecules, one of which is more prone to be protonated and the other prefers

to become the carbamate, are mixed, the formed carbamate will exhibit higher stability under CO₂ absorption conditions.

The reaction of CO₂ with aqueous solutions of piperidine and morpholine is monitored by FTIR and shown in Figure 3A,B. In the case of aqueous piperidine solution, at low amine conversion, the IR spectra exhibit absorption bands at 1510, 1430, 1460, and 1282 cm⁻¹. These can be assigned to the asymmetric and symmetric modes of C=O stretching of carbamate, N–H bending of NH₂⁺, and N–C stretching of NCOO⁻, respectively. At maximal piperidine conversion, all carbamate decomposes into bicarbonate, as signified by a strong 1360 cm⁻¹ band of C=O stretching from HCO₃⁻, 1010 cm⁻¹ of C–O stretching, and 950 cm⁻¹ of N–H twisting. These spectroscopic observations and assignments are consistent with previous studies.^{32,33,69} The spectral behaviors of morpholine and piperidine are quite similar. Carbamate is formed at low morpholine conversion, signified by the C=O asymmetric and symmetric stretching (1532 and 1425 cm⁻¹) and N–C stretching (1275 cm⁻¹). The carbamate product formed by morpholine also undergoes some decomposition as the 1530 cm⁻¹ band decreases, although this is less pronounced compared to piperidine. At equilibrium conversion, a significant amount of carbamate remains in the morpholine solution.

The amino site of piperidine has a pK_a of 11.2, while that of morpholine's amino site has a pK_a of 8.3.^{70,71} This significant difference in basic strength between the amino site of piperidine and that of morpholine suggests that when CO₂ reacts with a piperidine-morpholine mixture, piperidine will be protonated, while morpholine will hold the carbamate moiety. With this line of logic, the intersite stability concept predicts that the carbamate product formed from the mixed amine solution of piperidine and morpholine exhibits enhanced stability compared to the solution of a single amine, especially where the sole product at equilibrium is bicarbonate in the case of piperidine solution. To verify the prediction, reactions of CO₂ with mixed solutions of piperidine and morpholine at the same molar concentration of morpholine were conducted, and the results are shown in Figure 3C–E. Surprisingly, even at the beginning of the reaction, the carbamate moiety is only attached to the amino site of morpholine. This is evidenced by the strong and consistent presence of peaks of the asymmetric stretching mode of C=O at 1532 cm⁻¹ and the stretching mode of the N–C bond at 1275 cm⁻¹ from morpholine-COO⁻ throughout the reaction. In conjunction with the increase in the fingerprint band of morpholine-COO⁻, the 1460 and 950 cm⁻¹ bands, ascribed to the N–H bending of NH₂⁺ and N–H twisting of piperidine, are also well-defined, with no sign of bicarbonate being detected. These observations confirm the preferential roles of piperidine and morpholine when the mixed amine solution is reacted with CO₂. Furthermore, Figure 3E illustrates that starting with the same molar concentration of morpholine, the concentration of the carbamate moiety product attached to morpholine increases as more piperidine is added to the solution. Likewise, the equilibrium concentration of the carbamate moiety product formed by morpholine also increases following the incorporation of piperidine. These trends confirm the prediction from the concept of intersite stability. It is worth noting that mixing these amines does not entirely prevent the hydrolysis of carbamate into bicarbonate. This is because the intersite stability concept can only predict the direction of change of the carbamate stability as amino sites in the solution undergo some

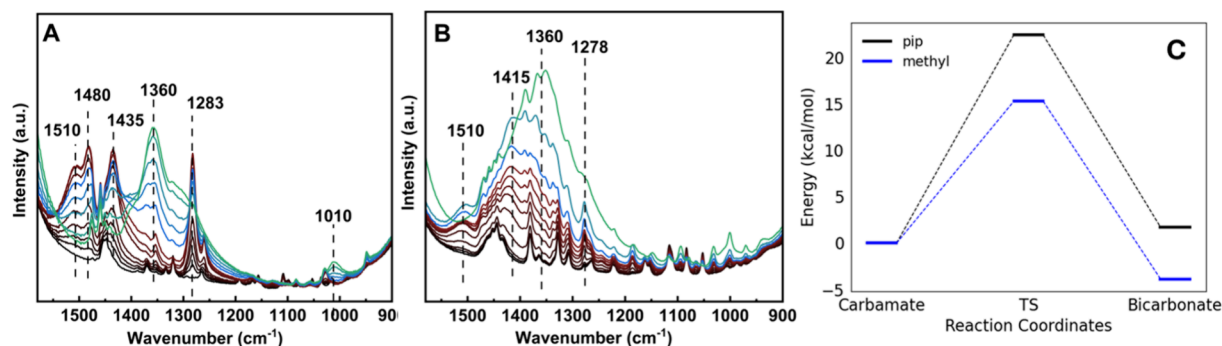


Figure 4. (A, B) FTIR spectra of the CO₂ capture reaction with aqueous 20 mol % piperidine (A) and aqueous 20 mol % 2-methyl-piperidine (B). (C) Activation energy barrier and reaction energy of carbamate hydrolysis for the CO₂ reaction with piperidine and 2-methylpiperidine solutions. The energy of each system is zeroed based on the respective carbamate as a reactant state. The activation energy barrier is the difference between the transition state (TS) and carbamate (reactant). The reaction energy is the difference between the bicarbonate (product) and carbamate (reactant). The black line represents piperidine carbamate + protonated piperidine, and the blue line represents 2-methylpiperidine carbamate + protonated 2-methylpiperidine. The exact values of these energies are listed in Table S2.

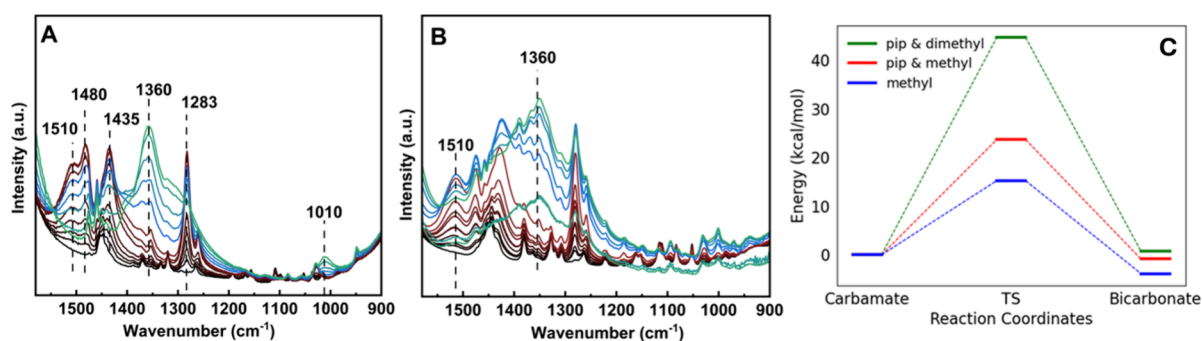


Figure 5. (A, B) FTIR spectra of the CO₂ capture reaction with aqueous 20 mol % piperidine (A) and an aqueous mixture of 20 mol % piperidine and 2-methyl-piperidine with the ratio of piperidine to 2-methyl-piperidine to be 2:1 (B). (C) Activation energy barrier and reaction energy of carbamate hydrolysis of piperidine and methyl-substituted piperidine. The energy of each system is zeroed based on the respective carbamate as the reactant state. The activation energy barrier is the difference between the transition state (TS) and carbamate (reactant). The reaction energy is the difference between the bicarbonate (product) and carbamate (reactant). The green line represents piperidine carbamate + protonated 2,6-dimethylpiperidine, the red line represents piperidine carbamate + protonated 2-methylpiperidine, and the blue line represents 2-methylpiperidine carbamate + protonated 2-methylpiperidine. The exact values of these energies are presented in Table S3.

chemical structure modifications. Our proposed concept does not aim to predict an amino structure that can completely prevent hydrolysis.

Similar to intramolecular modulation, the carbamate product stability can be influenced by factors beyond just the basicity of amino sites, such as steric hindrance. The effect of steric factors on the intersite stability of carbamate and the protonated site is highlighted through the comparison of piperidine and 2-methylpiperidine. Figure 4A,B shows the FTIR spectra of CO₂ absorption reactions in these amine solutions. The spectra illustrate that both piperidine and 2-methylpiperidine solutions react with CO₂ to form carbamate products at low CO₂ loading, which then decompose completely into bicarbonate products at equilibrium. However, the decomposition of carbamate of 2-methyl-piperidine happens at a much lower conversion level as compared to piperidine. This behavior has been observed in previous studies on cyclic amines.^{32,33} To gain a theoretical understanding of this behavior, DFT calculations were performed on the carbamate hydrolysis step for both solutions, and the results are presented in Figure 4C. The hydrolysis of 2-methyl-piperidine carbamate solution exhibits a negative reaction energy, while the hydrolysis of piperidine carbamate shows a slightly positive reaction energy. Furthermore, the activation energy of the carbamate hydrolysis reaction for piperidine is higher (7.14 kcal/mol) compared to

2-methyl-piperidine. This suggests that in the CO₂ absorption reaction, it is more spontaneous for 2-methyl-piperidine carbamate to be hydrolyzed compared to piperidine carbamate. Such a reaction tendency can be easily explained by invoking the concept of intersite stability. In detail, the methyl group at the alpha-position mainly imposes the steric effect with a weak induction on the amino site. Thus, the presence of this methyl group in 2-methyl-piperidine only slightly affects the protonated site while imposing significant steric hindrance that prevents the CO₂ binding on the amino site. Since both carbamate and protonated sites originate from the same structural unit (CHCH₃NHCH₂), protonation prefers the sterically hindered site, and carbamate formation prefers the less hindered site without a methyl group in the alpha-position. As a result, the carbamate site of (CHCH₃NCOO⁻CH₂) is less stable when being paired with (CHCH₃NH₂⁺CH₂) as the protonated site within the 2-methylpiperidine absorption solution.

In this vein, a simple modification, in light of intersite stability, can be applied to systematically reduce the decomposition of carbamate. In this case, the concept of intersite stability suggests that if piperidine is mixed with either 2-methyl-piperidine or 2,6-dimethyl-piperidine, the resulting solution will be more resistant to carbamate decomposition, at higher CO₂ absorption, when reacting with CO₂. A two-step

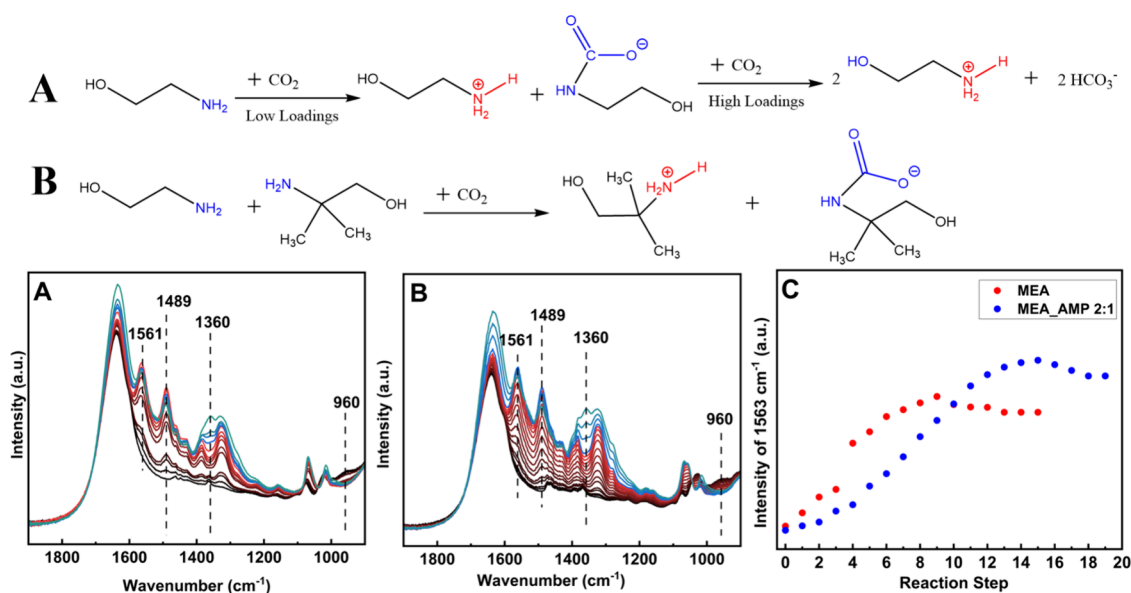


Figure 6. FTIR spectra of the CO₂ capture reaction with aqueous 20 mol % MEA (A) and an aqueous mixture of 20 mol % MEA and AMP with the ratio of MEA to AMP to be 2:1 (B) from 0% conversion to equilibrium conversion. (C) Intensity of carbamate C=O_{vas} (1561 cm⁻¹) peak extracted from A, B.

DFT calculation was performed to verify the prediction. In the first step, the carbamate formation reaction was investigated by considering whether the carbamate anion formed on piperidine and the proton bound to methyl-substituted piperidine, or vice versa. Figure S1 illustrates that in both cases, the carbamate formed on piperidine exhibits a more negative reaction energy and a slightly lower activation barrier compared to scenarios where the carbamate is formed on methyl-substituted piperidine.

Consequently, the theoretical calculation for the second step—the hydrolysis reaction of carbamate—only proceeded with the more favorable pairs of carbamate and proton moiety. In detail, Figure 5C illustrates the DFT calculations of carbamate hydrolysis for solutions containing (1) carbamate and protonated sites from only 2-methyl-piperidine, (2) a carbamate site held by piperidine and a protonated site held by 2-methyl-piperidine, (3) a carbamate site held by piperidine and a protonated site held by 2,6-dimethyl-piperidine. The reaction energy and activation energy of the hydrolysis reaction show an unequivocal increase in the order of (1) < (2) < (3). Thus, both reaction energy and activation energy converge in demonstrating that the carbamate stability increases as piperidine solution is introduced to methyl-substituted amine with a steric effect. This trend is completely in line with the intersite stability reasoning.

The enhancement of carbamate stability, induced by the steric hindrance, is also experimentally demonstrated by the observation in Figure 5A,B. The aqueous solution of only 2-methyl-piperidine can only form a low amount of carbamate before hydrolysis takes place and decomposes all carbamate products. When piperidine and 2-methyl-piperidine are mixed with the same concentration of 2-methyl-piperidine as in the previous experiment, the solution can capture a much higher amount of CO₂ as carbamate products, signified by well-defined peaks of C=O_{vas} at 1510 cm⁻¹, before the hydrolysis reaction takes over.

Even though the intersite stability of protonated and carbamate sites in this work has so far been demonstrated

on cyclic amine groups, this concept can very well be extended to other groups of amines, including small alkanol amines, which have drawn lots of interest for carbon capture commercialization. Figure 6 presents the effect of 2-amino-2-methyl-1-propanol (AMP) in addition to MEA on the carbamate stability. With AMP present in the solution, the reaction between MEA and CO₂ tends to form more carbamate, signified by the C=O_{vas} band at 1561 cm⁻¹, at both the maximum carbamate formation and equilibrium conversion. The sharp C=O_{vas} band at 1561 cm⁻¹ without any sign of a shoulder suggests the absence of an AMP-CO₂ carbamate moiety. It is worth mentioning that the product of CO₂ absorption in AMP aqueous solution has been experimentally and computationally confirmed.^{13,72} This implies that the increase in the carbamate concentration when AMP is added, shown in Figure 6C, is due to the increase of only the MEA-CO₂ carbamate moiety, even when the initial concentration of MEA is fixed. In other words, the addition of AMP promotes the stability of the MEA-CO₂ carbamate moiety by acting as a protonated site, AMPH⁺, which creates a more stable intersite interaction with MEA-CO₂ than the protonated MEAH⁺ site itself. The enhancement in stability in cyclic amines and alkanol amines, induced by intermolecular modulation, suppresses the formation of bicarbonate during various levels of CO₂ loading. This is very useful in designing amine blends and engineering molecule structures that improve the operational range of the absorption process in order to create a carbamate moiety as the unique absorption product and boost the reliability of predictive control and monitoring models. Last but not least, the blends of amines that become function-specific in reaction with CO₂ in some cases—blends of primary/secondary and tertiary amines—can also bring forth advantages in the kinetics. An example of this blend is the CANSOLV (DC-103), which has quickly become one of the dominant formulations. CANSOLV (DC-103) comprises piperazine, 1-piperazineethanol, and 1,4-piperazinediethanol. As compared to just piperazine, the CANSOLV (DC-103) formulation not

only forms a more stable carbamate, according to the intersite stability concept, due to the presence of tertiary amino sites, which are more prone to be protonated sites, but also is higher in reaction rate.⁷³ This demonstrates the possible application of the intersite stability in retrofitting the current industrial formulation targeting stable carbamate as well as other desirable engineering aspects.

CONCLUSIONS

This study introduces the novel concept of intersite stability as the key factor in controlling the stability of the carbamate products in amine solutions for CO₂ capture. Practically, this concept asserts that during CO₂ absorption, the stability of carbamate in amine solutions can be enhanced by deliberately assigning distinct roles to different amino sites through chemical or geometrical structural design: one site for protonation and the other for carbamate formation. As the differentiation between these roles becomes more pronounced, the stability of the carbamate increases. In other words, the more distinct and specialized the roles of these sites, the greater the stability of carbamate during the CO₂ absorption process. The intersite stability can be induced via either intramolecular or intermolecular site modulation and is applicable across a range of amine groups, including potentially commercialized amine blends, thereby improving the overall stability of the capture system. This concept marks the beginning of an innovative framework, paving the way for the development of more efficient and environmentally sustainable amine systems for carbon capture, thus advancing efforts to combat climate change. Future work should consider extending this investigation to a broader range of operational conditions, with direct comparisons to industrial parameters such as temperature, composition, and amine blend formulations.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsomega.5c03663>.

Optimized structures of amines and corresponding zwitterions and carbamates of piperidine, 2-methyl-piperidine, 2,6-methyl-piperidine, piperazine, 2,6-methyl-piperazine, 2,5-methyl-piperazine, and 1-methyl-piperazine; activation energy and reaction energy of piperidine carbamate and methyl-substituted piperidine carbamate; activation energy and reaction energy of carbamate formation of the solution of piperidine and methyl-substituted piperidine; and activation energy and reaction energy of hydrolysis reaction of 2,6-dimethyl-piperazine, 2,5-dimethyl-piperazine, and 1-methyl-piperazine (PDF)

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Notes

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REFERENCES

- (1) MacDowell, N.; Florin, N.; Buchard, A.; Hallett, J.; Galindo, A.; Jackson, G.; Adjiman, C. S.; Williams, C. K.; Shah, N.; Fennell, P. An Overview of CO₂ Capture Technologies. *Energy Environ. Sci.* **2010**, *3*, 1645–1669. November
- (2) Chai, S. Y. W.; Ngu, L. H.; How, B. S.; Chin, M. Y.; Abdouka, K.; Adini, M. J. B. A.; Kassim, A. M. Review of CO₂ Capture in Construction-Related Industry and Their Utilization. *International Journal of Greenhouse Gas Control* **2022**, *119*, No. 103727.
- (3) Heldebrant, D. J.; Koech, P. K.; Glezakou, V. A.; Rousseau, R.; Malhotra, D.; Cantu, D. C. Water-Lean Solvents for Post-Combustion CO₂ Capture: Fundamentals, Uncertainties, Opportunities, and Outlook. *Chem. Rev.* **2017**, *117*, 9594–9624. American Chemical Society July 26
- (4) D'Alessandro, D. M.; Smit, B.; Long, J. R. Carbon Dioxide Capture: Prospects for New Materials. *Angewandte Chemie - International Edition* **2010**, *49*, 6058–6082. August 16
- (5) Bara, J. E. What Chemicals Will We Need to Capture CO₂? *Greenh. Gases Sci. Technol.* **2012**, *2* (3), 162–171.
- (6) Maginn, E. J. *Ionic Liquids: Breakthrough Absorption Technology for Post-Combustion CO₂ Capture*; USDOE, 2013.
- (7) Park, Y.; Lin, K. Y. A.; Park, A. H. A.; Petit, C. Recent Advances in Anhydrous Solvents for CO₂ Capture: Ionic Liquids, Switchable Solvents, and Nanoparticle Organic Hybrid Materials. *Frontiers in Energy Research* **2015**. Frontiers Media S.A.
- (8) Budzianowski, W. M. Explorative Analysis of Advanced Solvent Processes for Energy Efficient Carbon Dioxide Capture by Gas-Liquid Absorption. *International Journal of Greenhouse Gas Control* **2016**, *49*, 108–120. Elsevier Ltd. June 1
- (9) Liang, Z.; Fu, K.; Idem, R.; Tontiwachwuthikul, P. Review on Current Advances, Future Challenges and Consideration Issues for Post-Combustion CO₂ Capture Using Amine-Based Absorbents. *Chinese Journal of Chemical Engineering* **2016**, *24*, 278–288. Chemical Industry Press February 1
- (10) Heldebrant, D. J.; Koech, P. K.; Rousseau, R.; Glezakou, V. A.; Cantu, D.; Malhotra, D.; Zheng, F.; Whyatt, G.; Freeman, C. J.; Bearden, M. D. Are Water-Lean Solvent Systems Viable for Post-Combustion CO₂ Capture? *Energy Procedia* **2017**, *114*, 756–763.
- (11) Ramdin, M.; De Loos, T. W.; Vlucht, T. J. H. State-of-the-Art of CO₂ Capture with Ionic Liquids. *Ind. Eng. Chem. Res.* **2012**, *51*, 8149–8177. June 20
- (12) Xie, H. B.; Zhou, Y.; Zhang, Y.; Johnson, J. K. Reaction Mechanism of Monoethanolamine with CO₂ in Aqueous Solution from Molecular Modeling. *J. Phys. Chem. A* **2010**, *114* (43), 11844–11852.
- (13) Yamada, H.; Matsuzaki, Y.; Higashii, T.; Kazama, S. Density Functional Theory Study on Carbon Dioxide Absorption into Aqueous Solutions of 2-Amino-2-Methyl-1-Propanol Using a Continuum Solvation Model. *J. Phys. Chem. A* **2011**, *115* (14), 3079–3086.

- (14) Matsuzaki, Y.; Yamada, H.; Chowdhury, F. A.; Higashii, T.; Onoda, M. Ab Initio Study of CO₂ Capture Mechanisms in Aqueous Monoethanolamine: Reaction Pathways for the Direct Interconversion of Carbamate and Bicarbonate. *J. Phys. Chem. A* **2013**, *117* (38), 9274–9281.
- (15) Mannisto, J. K.; Pavlovic, L.; Tiainen, T.; Nieger, M.; Sahari, A.; Hopmann, K. H.; Repo, T. Mechanistic Insights into Carbamate Formation from CO₂ and Amines: The Role of Guanidine-CO₂ adducts. *Catal. Sci. Technol.* **2021**, *11* (20), 6877–6886.
- (16) Conway, W.; Wang, X.; Fernandes, D.; Burns, R.; Lawrance, G.; Puxty, G.; Maeder, M. Comprehensive Kinetic and Thermodynamic Study of the Reactions of CO₂ (Aq) and HCO₃[−] with Monoethanolamine (MEA) in Aqueous Solution. *J. Phys. Chem. A* **2011**, *115* (50), 14340–14349.
- (17) da Silva, E. F.; Svendsen, H. F. Computational Chemistry Study of Reactions, Equilibrium and Kinetics of Chemical CO₂ Absorption. *Int. J. Greenh. Gas Control* **2007**, *1* (2), 151–157.
- (18) Mathias, P. M. Some Examples of the Contribution of Applied Thermodynamics to Post-Combustion CO₂-Capture Technology. *Fluid Phase Equilib.* **2014**, *362*, 102–107.
- (19) Mathias, P. M.; Reddy, S.; Smith, A.; Afshar, K. Thermodynamic Analysis of CO₂ Capture Solvents. *Int. J. Greenh. Gas Control* **2013**, *19*, 262–270.
- (20) Koech, P. K.; Zhang, J.; Kutnyakov, I. V.; Cosimbescu, L.; Lee, S. J.; Bowden, M. E.; Smurthwaite, T. D.; Heldebrant, D. J. Low Viscosity Alkanolguanidine and Alkanolamidine Liquids for CO₂ Capture. *RSC Adv.* **2013**, *3* (2), 566–572.
- (21) Freeman, S. A.; Rochelle, G. T. Density and Viscosity of Aqueous (Piperazine + Carbon Dioxide) Solutions. *J. Chem. Eng. Data* **2011**, *56* (3), 574–581.
- (22) Hunter, C. A.; Anderson, H. L. What Is Cooperativity? *Angew. Chemie Int. Ed.* **2009**, *48* (41), 7488–7499.
- (23) Forse, A. C.; Milner, P. J.; Lee, J.-H.; Redfearn, H. N.; Oktawiec, J.; Siegelman, R. L.; Martell, J. D.; Dinakar, B.; Zasada, L. B.; Gonzalez, M. I.; Neaton, J. B.; Long, J. R.; Reimer, J. A. Elucidating CO₂ Chemisorption in Diamine-Appended Metal-Organic Frameworks. *J. Am. Chem. Soc.* **2018**, *140* (51), 18016–18031.
- (24) Steinhardt, R.; Hiew, S. C.; Mohapatra, H.; Nguyen, D.; Oh, Z.; Truong, R.; Esser-Kahn, A. Cooperative CO₂ Absorption Isotherms from a Bifunctional Guanidine and Bifunctional Alcohol. *ACS Cent. Sci.* **2017**, *3* (12), 1271–1275.
- (25) Hampe, E. M.; Rudkevich, D. M. Exploring Reversible Reactions between CO₂ and Amines. *Tetrahedron* **2003**, *59* (48), 9619–9625.
- (26) Rudkevich, D. M.; Xu, H. Carbon Dioxide and Supramolecular Chemistry. *Chem. Commun.* **2005**, *21*, 2651–2659.
- (27) Masuda, K.; Ito, Y.; Horiguchi, M.; Fujita, H. Studies on the Solvent Dependence of the Carbamic Acid Formation from ω -(1-Naphthyl)Alkylamines and Carbon Dioxide. *Tetrahedron* **2005**, *61* (1), 213–229.
- (28) Dijkstra, Z. J.; Doornbos, A. R.; Weyten, H.; Ernsting, J. M.; Elsevier, C. J.; Keurentjes, J. T. F. Formation of Carbamic Acid in Organic Solvents and in Supercritical Carbon Dioxide. *J. Supercrit. Fluids* **2007**, *41* (1), 109–114.
- (29) Hwang, G. S.; Stowe, H. M.; Paek, E.; Manogaran, D. Reaction Mechanisms of Aqueous Monoethanolamine with Carbon Dioxide: A Combined Quantum Chemical and Molecular Dynamics Study. *Phys. Chem. Chem. Phys.* **2015**, *17* (2), 831–839.
- (30) Lv, B.; Guo, B.; Zhou, Z.; Jing, G. Mechanisms of CO₂ Capture into Monoethanolamine Solution with Different CO₂ Loading during the Absorption/Desorption Processes. *Environ. Sci. Technol.* **2015**, *49* (17), 10728–10735.
- (31) Versteeg, G. F.; van Swaaij, W. P. M. On the Kinetics between CO₂ and Alkanolamines Both in Aqueous and Non-Aqueous Solutions—I. Primary and Secondary Amines. *Chem. Eng. Sci.* **1988**, *43* (3), 573–585.
- (32) Robinson, K.; McCluskey, A.; Attalla, M. I. An ATR-FTIR Study on the Effect of Molecular Structural Variations on the CO₂ Absorption Characteristics of Heterocyclic Amines. *Part II. ChemPhysChem* **2012**, *13* (9), 2331–2341.
- (33) Robinson, K.; McCluskey, A.; Attalla, M. I. An FTIR Spectroscopic Study on the Effect of Molecular Structural Variations on the CO₂ Absorption Characteristics of Heterocyclic Amines. *ChemPhysChem* **2011**, *12* (6), 1088–1099.
- (34) Mohamed Mohsin, H.; Mohd Shariff, A.; Johari, K. 3-Dimethylaminopropylamine (DMAPA) Mixed with Glycine (GLY) as an Absorbent for Carbon Dioxide Capture and Subsequent Utilization. *Sep. Purif. Technol.* **2019**, *222*, 297–308.
- (35) Robinson, K.; McCluskey, A.; Attalla, M. I. Effect Molecular Structural Variations Has on the CO₂ Absorption Characteristics of Heterocyclic Amines. *Energy Procedia* **2012**, *11*, 224–231.
- (36) Heldebrant, D. J.; Koech, P. K.; Ang, M. T. C.; Liang, C.; Rainbolt, J. E.; Yonker, C. R.; Jessop, P. G. Reversible Zwitterionic Liquids, the Reaction of Alkanol Guanidines, Alkanol Amidines, and Diamines with CO₂. *Green Chem.* **2010**, *12* (4), 713–772.
- (37) Liu, A. H.; Ma, R.; Song, C.; Yang, Z. Z.; Yu, A.; Cai, Y.; He, L. N.; Zhao, Y. N.; Yu, B.; Song, Q. W. Equimolar CO₂ Capture by N-Substituted Amino Acid Salts and Subsequent Conversion. *Angew. Chemie - Int. Ed.* **2012**, *51* (45), 11306–11310.
- (38) Hong, S. Y.; Cheon, Y.; Shin, S. H.; Lee, H.; Cheong, M.; Kim, H. S. Carboxylate-Assisted Formation of Alkylcarbonate Species from CO₂ and Tetramethylammonium Salts with a β -Amino Acid Anion. *ChemSusChem* **2013**, *6* (5), 890–897.
- (39) Richner, G.; Puxty, G. Assessing the Chemical Speciation during CO₂ Absorption by Aqueous Amines Using in Situ FTIR. *Ind. Eng. Chem. Res.* **2012**, *51* (44), 14317–14324.
- (40) Im, J.; Hong, S. Y.; Cheon, Y.; Lee, J.; Lee, J. S.; Kim, H. S.; Cheong, M.; Park, H. Steric Hindrance-Induced Zwitterionic Carbonates from Alkanolamines and CO₂: Highly Efficient CO₂ Absorbents. *Energy Environ. Sci.* **2011**, *4* (10), 4284–4289.
- (41) O'Brien, M. J.; Farnum, R. L.; Perry, R. J.; Genovese, S. E. Secondary Amine Functional Disiloxanes as CO₂ Sorbents. *Energy Fuels* **2014**, *28* (5), 3326–3331.
- (42) Kumar, S.; Cho, J. H.; Moon, I. Ionic Liquid-Amine Blends and CO₂BOLs: Prospective Solvents for Natural Gas Sweetening and CO₂ Capture Technology-A Review. *International Journal of Greenhouse Gas Control* **2014**, *20*, 87–116. Elsevier Ltd.
- (43) Barzagli, F.; Mani, F.; Peruzzini, M. A Comparative Study of the CO₂ Absorption in Some Solvent-Free Alkanolamines and in Aqueous Monoethanolamine (MEA). *Environ. Sci. Technol.* **2016**, *50* (13), 7239–7246.
- (44) Ma, C.; Pietrucci, F.; Andreoni, W. Capturing CO₂ in Monoethanolamine (MEA) Aqueous Solutions: Fingerprints of Carbamate Formation Assessed with First-Principles Simulations. *J. Phys. Chem. Lett.* **2014**, *5* (10), 1672–1677.
- (45) Jackson, P.; Robinson, K.; Puxty, G.; Attalla, M. In Situ Fourier Transform-Infrared (FT-IR) Analysis of Carbon Dioxide Absorption and Desorption in Amine Solutions. *Energy Procedia* **2009**, *1*, 985–994.
- (46) Voice, A. K.; Vevelstad, S. J.; Chen, X.; Nguyen, T.; Rochelle, G. T. Aqueous 3-(Methylamino)Propylamine for CO₂ Capture. *Int. J. Greenh. Gas Control* **2013**, *15*, 70–77.
- (47) Neese, F. Software Update: The ORCA Program System—Version 5.0. *WIREs Comput. Mol. Sci.* **2022**, *12* (5), No. e1606.
- (48) Quapp, W.; Heidrich, D. Analysis of the Concept of Minimum Energy Path on the Potential Energy Surface of Chemically Reacting Systems. *Theor. Chim. Acta* **1984**, *66* (3–4), 245–260.
- (49) Sheppard, D.; Terrell, R.; Henkelman, G. Optimization Methods for Finding Minimum Energy Paths. *J. Chem. Phys.* **2008**, *128* (13), 134106.
- (50) Sheppard, D.; Henkelman, G. Paths to Which the Nudged Elastic Band Converges. *J. Comput. Chem.* **2011**, *32* (8), 1769–1771.
- (51) Ruttinger, A. W.; Sharma, D.; Clancy, P. Protocol for Directing Nudged Elastic Band Calculations to the Minimum Energy Pathway: Nurturing Errant Calculations Back to Convergence. *J. Chem. Theory Comput.* **2022**, *18* (5), 2993–3005.

- (52) Jonsson, H.; Mills, G.; Jacobsen, K. W. Nudged Elastic Band Method for Finding Minimum Energy Paths of Transitions. In *Classical and Quantum Dynamics in Condensed Phase Simulations*; WORLD SCIENTIFIC, 1998; pp 385–404.
- (53) Ásgeirsson, V.; Birgisson, B. O.; Björnsson, R.; Becker, U.; Neese, F.; Riplinger, C.; Jónsson, H. Nudged Elastic Band Method for Molecular Reactions Using Energy-Weighted Springs Combined with Eigenvector Following. *J. Chem. Theory Comput.* **2021**, *17* (8), 4929–4945.
- (54) Tomasi, J.; Mennucci, B.; Cammi, R. Quantum Mechanical Continuum Solvation Models. *Chem. Rev.* **2005**, *105* (8), 2999–3094.
- (55) Cossi, M.; Rega, N.; Scalmani, G.; Barone, V. Energies, Structures, and Electronic Properties of Molecules in Solution with the C-PCM Solvation Model. *J. Comput. Chem.* **2003**, *24* (6), 669–681.
- (56) Grimme, S.; Hansen, A.; Ehlert, S.; Mewes, J. M. R2SCAN-3c: A “Swiss Army Knife” Composite Electronic-Structure Method. *J. Chem. Phys.* **2021**, *154* (6), No. 064103.
- (57) Kruse, H.; Grimme, S. A Geometrical Correction for the Inter- and Intra-Molecular Basis Set Superposition Error in Hartree-Fock and Density Functional Theory Calculations for Large Systems. *J. Chem. Phys.* **2012**, *136* (15), 154101.
- (58) Caldeweyher, E.; Bannwarth, C.; Grimme, S. Extension of the D3 Dispersion Coefficient Model. *J. Chem. Phys.* **2017**, *147* (3), No. 034112.
- (59) Vydrov, O. A.; Van Voorhis, T. Nonlocal van Der Waals Density Functional: The Simpler the Better. *J. Chem. Phys.* **2010**, *133* (24), 244103.
- (60) Hujo, W.; Grimme, S. Performance of the van Der Waals Density Functional VV10 and (Hybrid)GGA Variants for Thermochemistry and Noncovalent Interactions. *J. Chem. Theory Comput.* **2011**, *7* (12), 3866–3871.
- (61) Weigend, F.; Ahlrichs, R. Balanced Basis Sets of Split Valence, Triple Zeta Valence and Quadruple Zeta Valence Quality for H to Rn: Design and Assessment of Accuracy. *Phys. Chem. Chem. Phys.* **2005**, *7* (18), 3297–3305.
- (62) Bui, T. Q.; Khokarale, S. G.; Shukla, S. K.; Mikkola, J.-P. Switchable Aqueous Pentaethylenhexamine System for CO₂ Capture: An Alternative Technology with Industrial Potential. *ACS Sustain. Chem. Eng.* **2018**, *6* (8), 10395–10407.
- (63) Kenarsari, S. D.; Yang, D.; Jiang, G.; Zhang, S.; Wang, J.; Russell, A. G.; Wei, Q.; Fan, M. Review of Recent Advances in Carbon Dioxide Separation and Capture. *RSC Adv.* **2013**, *3* (45), 22739–22773.
- (64) Said, R. B.; Kolle, J. M.; Essalah, K.; Tangour, B.; Sayari, A. A Unified Approach to CO₂–Amine Reaction Mechanisms. *ACS Omega* **2020**, *5* (40), 26125–26133.
- (65) Zanone, A.; Tavares, D. T.; Paiva, J. L. de An FTIR Spectroscopic Study and Quantification of 2-Amino-2-Methyl-1-Propanol, Piperazine and Absorbed Carbon Dioxide in Concentrated Aqueous Solutions. *Vib. Spectrosc.* **2018**, *99*, 156–161.
- (66) Kachko, A.; van der Ham, L. V.; Bardow, A.; Vlucht, T. J. H.; Goetheer, E. L. V. Comparison of Raman, NIR, and ATR FTIR Spectroscopy as Analytical Tools for in-Line Monitoring of CO₂ Concentration in an Amine Gas Treating Process. *Int. J. Greenh. Gas Control* **2016**, *47*, 17–24.
- (67) Andrews, J.; Dai, S. R.; Johnson, J. L. FTIR Spectra of Methyl-Substituted Amine-Hydrogen Fluoride Complexes in Solid Argon and Nitrogen. *J. Phys. Chem.* **1986**, *90*, 4273–4282.
- (68) Hong, S. Y.; Lee, J. S.; Cheong, M.; Kim, H. S. Isolation and Structural Characterization of Bicarbonate and Carbonate Species Formed during CO₂ Absorption/Desorption by a Hindered Alkanolamine. *Energy Procedia* **2014**, *63*, 2190–2198.
- (69) Einbu, A.; Ciftja, A. F.; Grimstvedt, A.; Zakeri, A.; Svendsen, H. F. Online Analysis of Amine Concentration and CO₂ Loading in MEA Solutions by ATR-FTIR Spectroscopy. *Energy Procedia* **2012**, *23*, 55–63.
- (70) Hartono, A.; Vevelstad, S. J.; Ciftja, A.; Knuutila, H. K. Screening of Strong Bicarbonate Forming Solvents for CO₂ Capture. *Int. J. Greenh. Gas Control* **2017**, *58*, 201–211.
- (71) Caplow, M. Kinetics of Carbamate Formation and Breakdown. *J. Am. Chem. Soc.* **1968**, *90* (24), 6795–6803.
- (72) Chakraborty, A. K.; Astarita, G.; Bischoff, K. B. CO₂ Absorption in Aqueous Solutions of Hindered Amines. *Chem. Eng. Sci.* **1986**, *41* (4), 997–1003.
- (73) Tozzi, M.; de Neufville, R. Flexible Design for Carbon Capture System Development: Economic Case of US Gulf Coast. In *Proceedings of the 17th Greenhouse Gas Control Technologies Conference (GHGT-17)*; SSRN, 2024.



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